Junior Topology Talk - Spin Structures Done Good

John Teague

10/23/2024

Abstract

In this talk, I will offer an intuitive approach to spin structures and Stiefel-Whitney classes in terms of Čech cohomology. My goal is that you will, once and for all, understand why Maggie says that a spin structure is “a trivialization of the 1-skeleton which extends over the 2-skeleton”, and why this is in direct analogy with a choice of orientation. I will mention the obstruction-theoretic approach but I'll mostly stick to low-tech explanations.

1 Vector Bundles

Intuitively, vector bundles are families of vector spaces parametrized by a particular base space (in our case, a manifold).

Definition 1. Let $\mathbb{K} = \mathbb{R}, \mathbb{C}$, or $\mathbb{H}$, V be an n-dimensional vector space over $\mathbb{K}$, and M be a smooth manifold of dimension m. Let $\{U_\alpha : \alpha \in A\}$ be an open covering of M, equipped with transition functions

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{GL}(V)$$

satisfying the “cocycle condition”

$$g_{\alpha\beta} \circ g_{\beta\gamma} = g_{\alpha\gamma} \text{ on } U_\alpha \cap U_\beta \cap U_\gamma.$$

Let $\widetilde{E}$ denote the set $A \times M \times V$, and define an equivalence relation on $\widetilde{E}$ by

$$(\alpha, p, v) \sim (\beta, q, w) \quad \text{if} \quad p = q \in U_\alpha \cap U_\beta, v = g_{\alpha\beta}(p)w.$$

Let $E = \widetilde{E}/\sim$, and define a projection map

$$\pi : E \to M \quad \text{by} \quad \pi([\alpha, p, v]) = p.$$

There is a unique smooth manifold structure on E whose charts are

$$\left\{\psi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times V\right\},$$

where

$$\psi_\alpha([\alpha, p, v]) = (p, v).$$

With respect to this structure, π is a smooth submersion.

A $\mathbb{K}$-*vector bundle of rank* n *over* M is a pair (E, π) constructed as above for some choice of open cover $\{U_\alpha : \alpha \in A\}$ of M and some collection $\{g_{\alpha\beta}\}_{\alpha,\beta\in A}$ of transition functions satisfying the cocycle condition.

Example 2.

(1) TM is a real vector bundle over M of rank m. We can build TM as above using an open cover: let $\{\varphi_\alpha : U_\alpha \to \mathbb{A}^n\}_{\alpha\in A}$ be the smooth atlas on M, where $\mathbb{A}^m$ is the n-dimensional affine space with transla-

tions $V = \mathbb{R}^m$. Then for the open cover $\{U_\alpha\}_{\alpha\in A}$, the transition functions

$$\{g_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{GL}(\mathbb{R}^n)\}$$

given by

$$g_{\alpha\beta}(p) = d_{\varphi_\alpha(p)}(\varphi_\beta \circ \varphi_\alpha^{-1})$$

satisfy the cocycle condition. It is straightforward to check that the real rank m vector bundle coming from this data is equivalent to your favorite definition of TM, and that the fiber over a point $p \in M$ is naturally identified with T_pM.

(2) The cotangent bundle T^*M can be constructed similarly, except with the transition functions being the dual maps

$$g_{\alpha\beta}(p) = d^*_{\varphi_\beta(p)}(\varphi_\beta \circ \varphi_\alpha^{-1}),$$

acting on the fibers which we identify naturally with T_p^*M.

(3) Similarly, one can explicitly construct the k-th exterior power $\Lambda^k T^*M$ of the cotangent bundle, as well as other tensor bundles. Given two vector bundles E_1, E_2 over M, one can form their direct sum $E_1 \oplus E_2$, their tensor product $E_1 \otimes E_2$, the Hom bundle $\mathrm{Hom}(E_1, E_2)$, etc.

(4) Let $E_1, \ldots, E_k$ be rank n vector bundles over M. For any (continuous) functor

$$F : \bigtimes_k \mathrm{Vect}_n^{\mathbb{R}} \to \mathrm{Vect}_n^{\mathbb{R}}$$

where $\mathrm{Vect}_n^{\mathbb{R}}$ is the category whose objects are n-dimensional vector spaces over $\mathbb{R}$ and whose morphisms are isomorphisms, we can construct a new bundle whose fiber over a point p is $T((E_1)_p, \ldots, (E_k)_p)$. This general statement subsumes the above examples.[1]

Definition 3. Let $\pi : E \to M$ and $\pi' : E' \to M$ be vector bundles over M. A morphism of vector bundles $F : (\pi, M) \to (\pi', M)$ is a map $F : E \to E'$ such that the diagram

$$\begin{tikzcd} E \arrow[rr, "F"] \arrow[dr, "\pi"'] & & E' \arrow[dl, "\pi'"] \\ & M & \end{tikzcd}$$

commutes. That is, the induced maps $F_p : E_p \to E'_p$ are linear for all $p \in M$.

Remark. Note that $\mathrm{GL}(\mathbb{H}^n)$ is a subgroup of $\mathrm{GL}(\mathbb{C}^{2n})$, which is in turn a subgroup of $\mathrm{GL}(\mathbb{R}^{4n})$. As such, we can think of any quaternionic vector bundle of rank n as a real vector bundle of rank $4n$ whose transition functions take values in $\mathrm{GL}(\mathbb{H}^n) < \mathrm{GL}(\mathbb{R}^{4n})$.

More generally, if G is any Lie subgroup of $\mathrm{GL}(\mathbb{R}^n)$, a G-vector bundle is a rank n vector bundle whose transition functions take their values in G.

Example 4. *(a) Consider $G = \mathrm{GL}^+(\mathbb{R}^n) < \mathrm{GL}(\mathbb{R}^n)$, the connected subgroup of $\mathrm{GL}(\mathbb{R}^n)$ containing the identity. G consists of maps with positive determinant, or equivalently those which preserve orientation. As such, this yields an orientation on such a G-vector bundle.*

(b) Consider $G = O(n) < \mathrm{GL}(\mathbb{R}^n)$. In this case, the transition functions of a G-vector bundle preserve the usual dot product on $\mathbb{R}^n$. Thus, we obtain a natural fiber metric given by pushing forward the dot product along the transition functions. If G is further restricted to $SO(n)$, M inherits an orientation as well.

(c) If $G = U(n) \subset \mathrm{GL}(\mathbb{C}^n) < \mathrm{GL}(\mathbb{R}^{2n})$, a G-vector bundle is a complex vector bundle of rank n together with a Hermitian metric.

[1] See Milnor-Stasheff for more.

Definition 5. In general, if we're able to construct the tangent bundle TM such that the transition maps between trivializing charts lie in G, we say that M has a *G-structure*.

The examples from above show that this includes things such as, orientation, Riemannian metrics, Hermitian metrics (hence complex structures), and many more (including spin structures!).

2 Obstructing Orientations

This section is going to be *very* hand-wavy and impressionistic, but we will formalize these thoughts later when we have the language to do so.

Definition 6. An open cover $\{U_\alpha\}$ of a manifold M is a *good cover* if

$$U_{\alpha_1} \cap U_{\alpha_2} \cap \cdots \cap U_{\alpha_k}$$

is either empty or contractible for all possible $(\alpha_1, \ldots, \alpha_k)$.

Suppose we choose a good cover $\{U_\alpha\}$ as a trivializing cover for TM, with transition maps $g_{\alpha\beta}$. It's always possible to require the transition functions to take their values in $\mathrm{O}(m)$, since $\mathrm{GL}(\mathbb{R}^m)$ deformation retracts onto this subgroup (hence every manifold admits a Riemannian metric). All we need to do in order to choose an orientation on M (technically, an orientation on TM) is to make a choice of orientation on each U_α such that the transition maps preserve orientation. Let's do this by hand for the real line bundles over S^1.

Consider the standard stereographic charts on S^1 given by $U_N = S^1 \setminus \{N\}$ and $U_S = S^1 \setminus \{S\}$ where N and S are the north and south pole of S^1. Then we have the transition map

$$g_{NS} : U_N \cap U_S = S^1 \setminus \{U, N\} \to \mathrm{O}(\mathbb{R}) = \{-1, +1\}\,.$$

Notice that, since $U_N \cap U_S$ is the union of two disconnected contractible subsets of S^1, the data of g_{NS} is exactly a choice of (multiplication by) $+1$ or -1 on either component; that is, a choice of whether to preserve or reverse orientation as we move over either intersection.

(1) First, consider $TS^1 \cong S^1 \times \mathbb{R}$, which is a cylinder. The transition map g_{NS} defines the trivial bundle over S^1 if and only if the product over all components of the intersection is $+1$, i.e. either acting by 1 and 1 or -1 and -1 with the trivializing charts we've chosen. Let's construct an orientation for either of these choices by hand. There is some subtlety to what this actually means; although there are exactly two orientations on either component, they are not in canonical bijection with $\{-1, +1\}$. Our strategy will be to make a choice of orientation on one of the trivializing charts which we will call "1" and attempt to propagate it over the rest of the charts using the given transition maps.

   (a) In the case where g_{NS} acts by the identity on either component, this is rather trivial. Choose an orientation on $U_N \times \mathbb{R}$ and pick the "same" orientation on $U_S \times \mathbb{R}$, where "same" means the "image" of this orientation along the identity.

   (b) In the case where g_{NS} acts by -1 on both components, fix an orientation on U_N and take the "opposite" orientation on U_S.

(2) But what about the case where we know there isn't a consistent choice of orientation — how do we measure this? Let's repeat the same process as above for the (open) Möbius band, with the usual projection onto S^1. This is given by transition maps acting by ± 1 on either component. Choose an orientation on U_N. If we move over the orientation-preserving transition map, we choose the "same" orientation on U_S. But then the other transition map flips to the opposite orientation over U_N. Similarly, if we first move over the orientation-reversing transition map and choose the "opposite" orientation on U_S, we run into the same issue.

A priori, our obstruction seems to require a *lot* of data. We need to try every possible propagation along every possible transition map. As it turns out, the obstruction above boiled down to a single number in $\{-1, +1\}$, the product of all of the transition maps. This can be encoded very naturally as a $\mathbb{Z}/2$ Čech cohomology class.

3 Čech Cohomology

Čech cohomology is a cohomology theory associated to good covers of nice spaces (e.g. manifolds) — it is the simplicial cohomology of the nerve $N(\mathcal{U})$ associated to a good cover $\mathcal{U}$. As such, it is a natural way to encode the obstructions to putting various structures on vector bundles. In particular, the "cocycle condition" form above will allow us to succinctly encode these obstructions as certain $\mathbb{Z}/2$ Čech cohomology classes, which, as it turns out, are naturally identified with classes in ordinary singular cohomology. I won't go into detail on why this is the case, but for the rest of this talk you should have in mind that the obstruction classes we talk about live in singular cohomology.

Let X be a manifold and $\mathcal{U}$ a good cover of X.

Definition 7. A Čech n-simplex σ of $\mathcal{U}$ is an ordered collection of $n+1$ sets in $\mathcal{U}$ such that the intersection of all of these sets is nonempty.

Let $\sigma = \{U_i\}$ be such an n-simplex. The *j-th partial boundary* of σ is the $(n-1)$-simplex obtained by removing the j-th set from σ:

$$\partial_j \sigma \coloneqq \left\{U_1, \ldots, \widehat{U_j}, \ldots, U_n\right\}.$$

The *boundary* of σ is the alternating sums of the partial boundaries:

$$\partial\sigma \coloneqq \sum_{j=0}^{n} (-1)^{j+1} \partial_j \sigma.$$

Definition 8. An Čech n-cochain of $\mathcal{U}$ with coefficients in $\mathbb{Z}/2$ is a map which associates to each n-simplex σ and element of $\mathbb{Z}/2$, and we denote the set of all $\mathbb{Z}/2$ n-cochains by $C^n(\mathcal{U}, \mathbb{Z}/2)$. These are all abelian groups with pointwise addition.

The *Čech cochain complex* is the cochain complex $C^*(\mathcal{U}, \mathbb{Z}/2)$ with coboundary operator $\delta_n : C^n(\mathcal{U}, \mathbb{Z}/2) \to C^{n+1}(\mathcal{U}, \mathbb{Z}/2)$ given by[2]

$$(\delta_n f)(\sigma) \coloneqq \sum_{j=0}^{n+1} (-1)^j f(\partial_j \sigma).$$

By design, we have that $\delta_{n+1} \circ \delta_n = 0$. We thus have our ordinary definitions of *Čech cocycle*, the n-cochains in the kernel of δ_n, and *Čech coboundaries*, n-cochains in the image of δ_{n-1}.

The *Čech cohomology of X with $\mathbb{Z}/2$ coefficients* $\check{H}^*(X, \mathbb{Z}/2)$ is the cohomology of this cochain complex.

Theorem 9. *The Čech cohomology of X is isomorphic to singular cohomology of X:*

$$\check{H}^*(X, \mathbb{Z}/2) \cong H^*(X, \mathbb{Z}/2).$$

In general, this is true for any topological space which is homotopy equivalent to a CW complex (and admits a good cover, for our definition to work).

We can now define the first Stiefel-Whitney class of an n-manifold M. Choose a good cover $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ as a trivializing cover for TM, and let the transition functions take values in $O(n)$:

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \to O(n).$$

[2]Ordinarily, Čech cohomology can be valued in any *presheaf* $\mathcal{F}$ on X, and this formula would require factoring f through the restriction morphism from $\mathcal{F}(|\partial_j\sigma|)$ to $\mathcal{F}(|\sigma|)$. In this more general setting, "$\mathbb{Z}/2$ coefficients" indicates taking $\mathcal{F}$ to be the constant sheaf determined by $\mathbb{Z}/2$.

Let $\eta_{\alpha\beta} = \det(g_{\alpha\beta}) \in \{\pm 1\}$. The set $\eta = \{\eta_{\alpha\beta}\}_{\alpha,\beta\in A}$ is exactly the data of a Čech cochain, a choice of element of $\mathbb{Z}/2$ in the support of every 1-simplex. Moreover, because transition functions satisfy the cocycle condition, η is a Čech cocycle, and hence represents an element of $\check{H}^1(M, \mathbb{Z}/2)$.

Definition 10. The first Stiefel-Whitney class of M, denoted $w_1(TM)$, is the cohomology class represented by η.

Theorem 11. *M admits an orientation if and only if $w_1(TM)$ vanishes.*

Proof. If M has an orientation, we can choose all of the transition maps to have determinant 1, hence $w_1(TM)$ is zero. Conversely, if $w_1(TM)$ is zero, then η must be a Čech coboundary. This implies exactly that there exists maps $\eta_\alpha : U_\alpha \to \{\pm 1\}$ such that $\eta_{\alpha\beta} = \eta_\alpha \eta_\beta^{-1}$. Applying this map to every trivializing chart guarantees that each transition map has determinant 1, hence M has an orientation. ☐

4 Spin Structures

Now we're ready to explore spin structures. First, a bit of motivation for why you should care. I'm sure you've seen these come up naturally in your life, but let me give a couple of results which illustrate how the existence of a spin structure can influence the topology of a manifold:

Theorem 12. *(Rokhlin) If X is a smooth, closed, spin 4-manifold (that is, admits a spin structure), then* $\sigma(X) = 0 \pmod{16}$.

Example 13.

(1) A complex surface in $\mathbb{CP}^3$ of degree d is spin if and only if d is even.

(2) A K3 surface admits a spin structure (being a complex projective surface of degree 4), and has signature -16, hence 16 is the best possible number in Rokhlin's theorem.

(3) Freedman's E_8 manifold is a simply connected, compact topological manifold with a spin structure, but whose intersection form is signature 8. Rokhlin's theorem thus does not hold for non-smooth manifolds in general, and proves that E_8 doesn't admit a smooth structure.

Spin structures also play a central role in *gauge theory*, which was a source of major innovations in low-dimensional topology over the last 40-50 years or so.

Ok, but what is a spin structure? Before we answer this question, we need to talk about $\mathrm{Spin}(n)$. There are a number of ways to define $\mathrm{Spin}(n)$, but the one most useful to us is as follows

Definition 14. As a manifold, $\mathrm{Spin}(n)$ is the double cover of $\mathrm{SO}(n)$. Then, as you may have learned in Tim's class, we can lift the Lie group structure of $\mathrm{SO}(n)$ to $\mathrm{Spin}(n)$ by choosing a lift of the identity.

Example 15.

(1) Since $\mathrm{SO}(3) \cong \mathbb{RP}^3$, *we have that* $\mathrm{Spin}(3) \cong S^3 \cong \mathrm{SU}(2)$.

(2) It turns out that $\mathrm{Spin}(4) \cong \mathrm{SU}(2) \times \mathrm{SU}(2)$.

Let M be an oriented manifold. Because M is orientable, we can choose a trivializing open cover $\mathcal{U} = \{U_\alpha\}_{\alpha\in A}$ for TM so that the corresponding transition functions take their values in $\mathrm{SO}(n)$:

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \to SO(n) \subset \mathrm{GL}(\mathbb{R}^4).$$

Let $\rho : \mathrm{Spin}(n) \to \mathrm{SO}(n)$ be the two-to-one covering map.

Definition 16. A *spin structure* on M is given by an open covering $\mathcal{U}$ of M and a collection of transition functions

$$\widetilde{g}_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{Spin}(n)$$

such that $\rho \circ \widetilde{g}_{\alpha\beta} = g_{\alpha\beta} \in \mathrm{SO}(n)$ and the cocycle condition

$$\widetilde{g}_{\alpha\beta}\widetilde{g}_{\beta\gamma} = \widetilde{g}_{\alpha\gamma}$$

on $U_\alpha \cap U_\beta \cap U_\gamma$ is satisfied. In other words, a spin structure on M is a *lift* of all of the transition functions from $\mathrm{SO}(n)$ to $\mathrm{Spin}(n)$. Manifolds which admit spin structures are called *spin manifolds*.

Let $\mathcal{U}$ be a good cover which is a trivializing cover for TM. Because $U_\alpha \cap U_\beta$ is contractible for all $\alpha, \beta \in A$, we can lift each

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{SO}(n) \quad \text{to} \quad \widetilde{g}_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{Spin}(n).$$

The only problem is that the cocycle condition may not hold. However, in a very similar way to how we obstructed orientations, we can define an invariant which exactly determines whether this condition can be enforced. For any $\alpha, \beta, \gamma \in A$, we can define

$$\eta_{\alpha\beta\gamma} = \widetilde{g}_{\alpha\beta}\widetilde{g}_{\beta\gamma}\widetilde{g}_{\gamma\alpha} : U_\alpha \cap U_\beta \cap U_\gamma \to \{\pm 1\} = \mathbb{Z}/2.$$

This is valued in $\{\pm 1\}$ because it is a lift of the identity in $\mathrm{SO}(n)$, and one can show that this implies either the identity or -1. As such, the data $\eta = \{\eta_{\alpha\beta\gamma}\}_{\alpha,\beta,\gamma\in A}$ is exactly a Čech 2-cochain of $\mathcal{U}$ with coefficients in $\mathbb{Z}/2$. Moreover, I claim that it is a cocycle — let σ be a 3-simplex in $\mathcal{U}$. σ consists of 4 sets $\{U_\alpha, U_\beta, U_\gamma, U_\delta\}$ in $\mathcal{U}$ with nonempty intersection, and so we can consider the η gotten by choosing 3 of these 4 indices. We compute

$$(\delta\eta)(\sigma) = \eta_{\beta\gamma\delta}\eta^{-1}_{\alpha\gamma\delta}\eta_{\alpha\beta\delta}\eta^{-1}_{\alpha\beta\gamma} = 1,$$

and so $\delta\eta$ vanishes on any 3-simplex, hence η is a Čech cocycle. As such, η represents a cohomology class in $\check{H}^2(M; \mathbb{Z}/2)$, and we will call it the *second Stiefel-Whitney class*, and denote it $w_2(TM)$.

Theorem 17. *M admits a spin structure if and only if $w_2(TM) = 0$.*

Proof. If M has a spin structure, we can choose the $\widetilde{g}_{\alpha\beta}$'s to satisfy the cocycle condition, so the Stiefel-Whitney class must be 0. Conversely, if the second Stiefel-Whitney class is zero, Čech cohomology tells us that it is a coboundary, and so there exists constant maps

$$\eta_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathbb{Z}/2 \quad \text{such that} \quad \eta_{\alpha\beta}\eta_{\beta\gamma}\eta_{\gamma\alpha} = \eta_{\alpha\beta\gamma}.$$

Then $\{\eta_{\alpha\beta}\widetilde{g}_{\alpha\beta}\}$ are transition functions which define a spin structure on M. ☐

Theorem 18. *If the Stiefel-Whitney class of index i is nonzero, then there cannot exist $(n - i + 1)$ everywhere linearly independent sections of the vector bundle.*

Concluding Thoughts

These obstructions are part of the general story of obstruction theory. This involves cohomology groups with coefficients in homotopy groups to define obstructions to extensions. For example, with a mapping from a simplicial complex X to another, Y, defined initially on the 0-skeleton of X (the vertices of X), an extension to the 1-skeleton will be possible whenever the image of the 0-skeleton will belong to the same path-connected component of Y. Extending from the 1-skeleton to the 2-skeleton means defining the mapping on each solid triangle from X, given the mapping already defined on its boundary edges. Likewise, then extending the mapping to the 3-skeleton involves extending the mapping to each solid 3-simplex of X, given the mapping already defined on its boundary.

At some point, say extending the mapping from the (n-1)-skeleton of X to the n-skeleton of X, this procedure might be impossible. In that case, one can assign to each n-simplex the homotopy class $\pi_{n-1}(Y)$ of the mapping already defined on its boundary, (at least one of which will be non-zero). These assignments define an n-cochain with coefficients in $\pi_{n-1}(Y)$. Amazingly, this cochain turns out to be a cocycle and so defines a cohomology class in the nth cohomology group of X with coefficients in $\pi_{n-1}(Y)$. When this cohomology class is equal to 0, it turns out that the mapping may be modified within its homotopy class on the (n-1)-skeleton of X so that the mapping may be extended to the n-skeleton of X. If the class is not equal to zero, it is called the obstruction to extending the mapping over the n-skeleton, given its homotopy class on the (n-1)-skeleton.

In particular, this framework can be used to extend sections of principle bundles. Suppose that B is a simply connected simplicial complex and that $p : E \to B$ is a fibration with fiber F. Furthermore, assume that we have a partially defined section $\sigma_n : B_n \to E$ on the n-skeleton on B.

For every $(n+1)$-simplex Δ in B, σ_n can be restricted to the boundary $\partial\Delta$ (an n-sphere). Because p sends each $\sigma_n(\partial\Delta)$ back to $\partial\Delta$, σ_n defines a map from the n-sphere to $p^{-1}(\Delta)$. Because fibrations satisfy the homotopy lifting property and Δ is contractible, $p^{-1}(\Delta)$ is homotopy equivalent to F. So, this partially defined section assigns an element of $\pi_n(F)$ to every $(n+1)$-simplex. This is precisely the data of a $\pi_n(F)$-valued simplicial cochain of degree $n+1$ on B, i.e. an element of $C^{n+1}(B; \pi_n(F))$. This cochain is called the *obstruction cochain* because it being zero means that all of these elements of $\pi_n(F)$ are trivial, which means that our partially defined section can be extended to the $(n+1)$-skeleton using the homotopy between (the partially defined section on the boundary of each Δ) and the constant map.